YEAR 7 MATHS · CHAPTER 1 · MIND MAP

Computation with positive integers

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1 Ancient number systems

Roman numerals add up, but a smaller symbol before a larger one means subtract.

I = 1, V = 5, X = 10, L = 50, C = 100

XIV = 14 XL = 40 CXXIII = 123

2 Place value

Each digit's value depends on where it sits. A zero holds an empty place.

The 3 in 4325 is worth 300

$4325 = 4 \times 1000 + 3 \times 100 + 2 \times 10 + 5$

507: the zero holds the tens place

3 Adding and subtracting

Round to a friendly number, then adjust.

$47 + 28 = 47 + 30 - 2 = 75$

$63 - 18 = 63 - 20 + 2 = 45$

Order doesn't matter: $38 + 17 + 2 = 38 + 2 + 17 = 57$

4 Algorithms for + and −

Line up the place values, then carry or borrow.

$476 + 385 = 861$

$603 - 247 = 356$

Check: $356 + 247 = 603$ ✓

Watch out: line up the units, not the left-hand digits.

5 Multiplying small numbers

Split a number into tens and units. Order doesn't matter, so pair numbers that are easy.

$6 \times 23 = 6 \times 20 + 6 \times 3 = 138$

$4 \times 17 \times 25 = 100 \times 17 = 1700$

$5 \times 14 \times 2 = 10 \times 14 = 140$

6 Multiplying large numbers

Multiply by each digit, then add the parts.

$37 \times 24 = 37 \times 20 + 37 \times 4 = 740 + 148 = 888$

$45 \times 100 = 4500$

$123 \times 6 = 738$

7 Dividing

Short division: divide one digit at a time, carrying what's left.

$84 \div 4 = 21$ $936 \div 8 = 117$

$47 \div 5 = 9$ remainder 2

Check: $117 \times 8 = 936$ ✓

8 Estimating and rounding

Look at the next digit: 5 or more rounds up.

387 to the nearest 10 is **390**, to the nearest 100 is **400**

Estimate $48 \times 21 \approx 50 \times 20 = 1000$

Estimate $612 \div 3 \approx 600 \div 3 = 200$

9 Order of operations

Brackets first, then × and ÷ from left to right, then + and − from left to right.

$3 + 4 \times 5 = 23$ $(3 + 4) \times 5 = 35$

$20 - 12 \div 4 = 17$

$2 \times (6 + 4) - 5 = 15$

Watch out: don't just work from left to right.